

Introduction to Generative AI

Generative AI

Module 1 – Lesson 1

“Generative models are a key enabler of machine creativity, allowing machines to go beyond what they’ve seen before and create something new”

- Ian Goodfellow, Computer Scientist

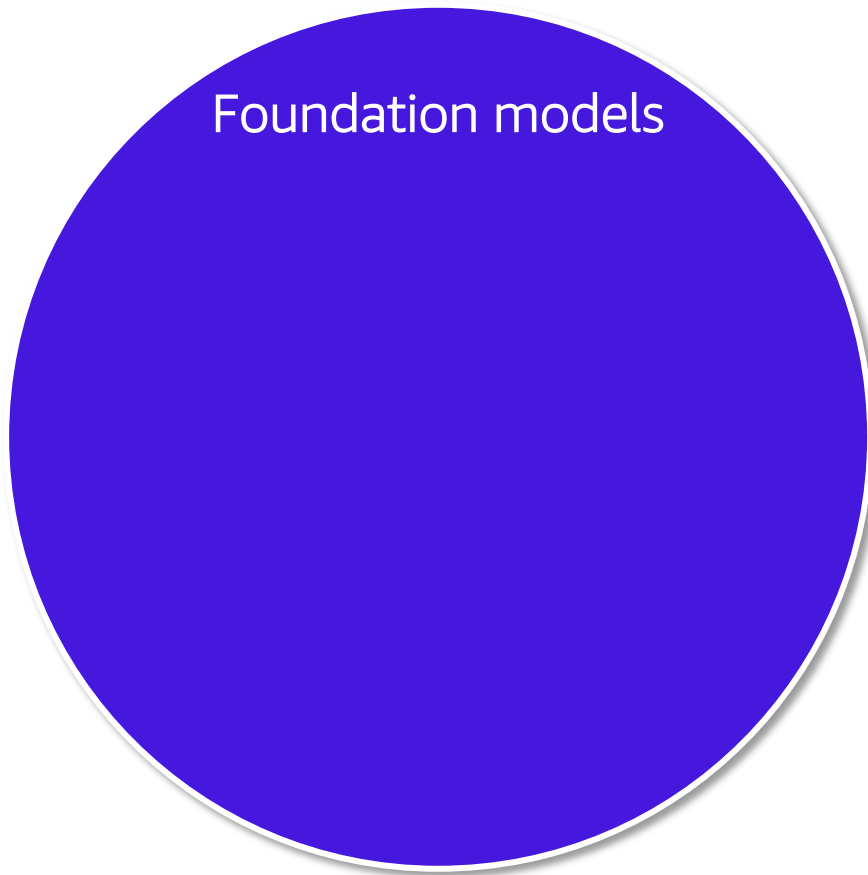
Today's activities

- Foundation models and LLMs
- Use cases of LLMs
- Amazon Bedrock



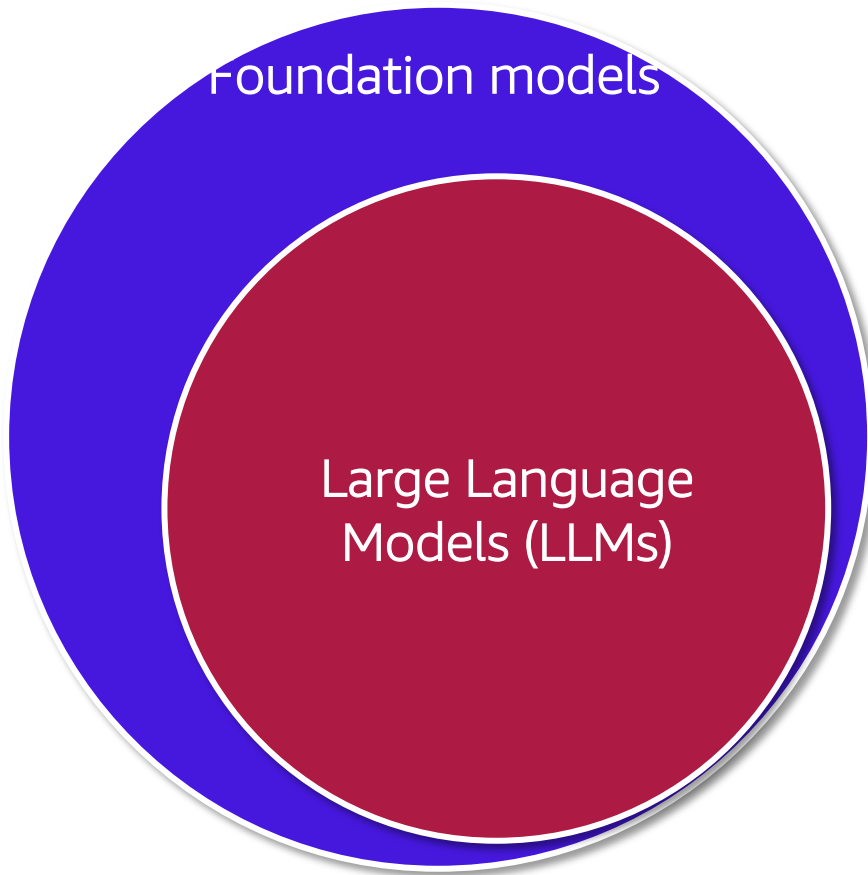
Introduction to generative AI

Foundation models



- Large ML models that are **pre-trained** with **vast amounts of data**. These can be **adapted** to more specialized tasks
- Can be trained on any kind of data
 - Text
 - Images
 - Video
 - Audio
 - Etc...

Large Language Models (LLMs)



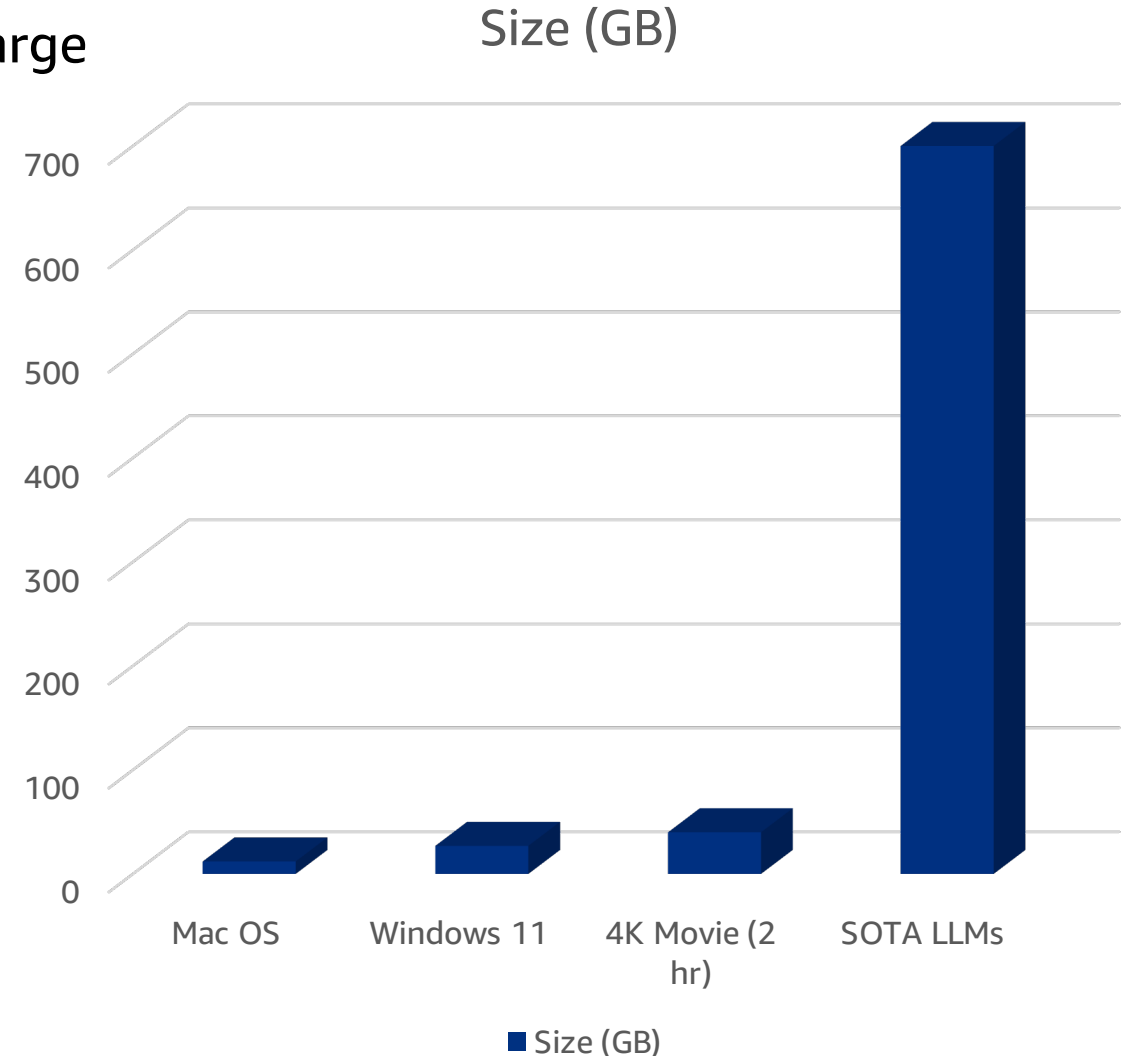
- Foundation models trained on **text**
- Large ML models that learn the **probabilities of words** being used in certain contexts
- **Training task:** Learn to predict the missing word in a text sequence

"The weather has been cloudy for the last two days. Most likely it will be ____ tomorrow."

cloudy? sunny? foggy?

How big are foundation models?

- Size of state-of-the-art (SOTA) models are as large as:
 - 474,000,000-page document
 - 35 hours of 4K content
 - Codebase with 80,000,000,000 lines of code
- Requires a lot of resources
 - Can cost more than \$100 million
 - Hundreds of people involved



Use cases of LLMs

Revolutionizing various domains



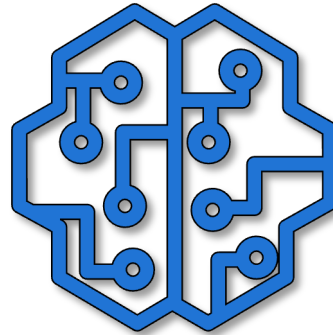
Healthcare



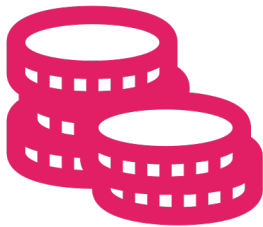
Education



Customer Service



FMs



Finance



Law



Marketing

Conversational chatbots

- Interactive chat applications
- Human-like dialogues
- Personalized responses
- Conversational awareness
 - Allow follow-up questions
- Used as powerful virtual assistants

Interactive training

- Rapid content generation and adaptation
- Dynamic and personalized content
- Accessibility and inclusivity for diverse audiences
- Multilingual support
- Create slides, exercises, quizzes, explanations for specific use cases

Creative assistant

- Creative content generation
- Intuitive prompt-based guidance:
 - Generate artistic works
 - Generate music
 - Generate written content
- Adapting content using input prompts, images, audio, etc.

Productivity tools

- Automate routine, trivial tasks
- Document writing
 - Generate drafts
 - Format, edit, summarize documents
- Code generation
 - Implement features and functionality
 - Code formatting, commenting and restructuring
 - Test case writing
- Efficient communication
 - Draft, summarize and auto-complete emails
 - Personalize responses for different groups, teams, individuals, etc.

Data analytics

- Uncover hidden patterns from data
 - Sentiments, PII, topics, etc
- Analyze charts, graphs and other visual data
- Generate insightful reports
 - Suggest potential solutions
- Create synthetic data for testing and training

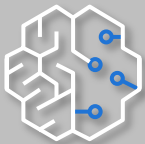
Amazon Bedrock

Amazon Bedrock

A fully-managed service that makes **foundation models** available via an API.

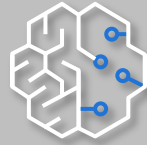
Some foundation models (FMs) available:

Amazon



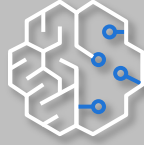
Titan

AI21 Labs



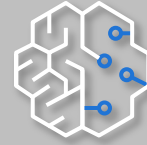
Jurassic-2

Anthropic



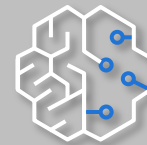
Claude

Cohere



Command

Meta

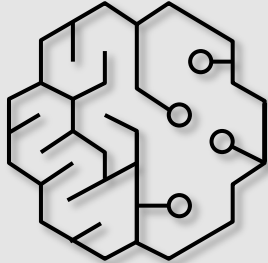


Llama

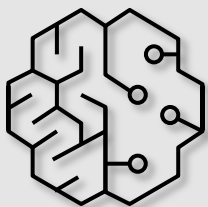
Amazon Bedrock

- Customers can:
 - **Privately customize FMs** with their own data
 - Easily **integrate them into their applications** using AWS tools and capabilities without having to provision or manage any infrastructure
- Prompts and responses are **not shared** with AWS or third-party providers
- Bedrock provides **additional security capabilities** such as encryption, identity and access management (IAM), and various compliance designations

Amazon Titan models

Amazon Titan	Benefits:
Innovate responsibly with high-performing FMs from Amazon.  Titan models	⚙ Automate natural language tasks such as summarization and text generation ⚙ Enhance search and improve personalized recommendations with Titan Embeddings ⚙ Support responsible use of AI by reducing inappropriate or harmful content

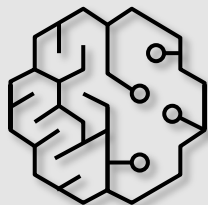
Amazon Titan Text Models



Titan Text

A generative foundation model for tasks such as:

- ✿ Summarization
- ✿ Text generation (for example, creating a blog post)
- ✿ Classification
- ✿ Open-ended Q&A
- ✿ Information extraction.



**Titan
Embeddings**

Translates text inputs (words, phrases, or possibly large units of text) into numerical representations (known as embeddings) that contain the semantic meaning of the text.

Bedrock Use Cases (1/2)



Text generation: Create new pieces of original content, such as short stories, essays, social media posts, and webpage.



Chatbots: Build conversational interfaces such as chatbots and virtual assistants to enhance the user experience for your customers.



Search: Search, find, and synthesize information to answer questions from a large corpus of data.

Bedrock Use Cases (2/2)



Text summarization: Get a summary of textual content, such as articles, blog posts, books, and documents, to get the gist without having to read the full content.



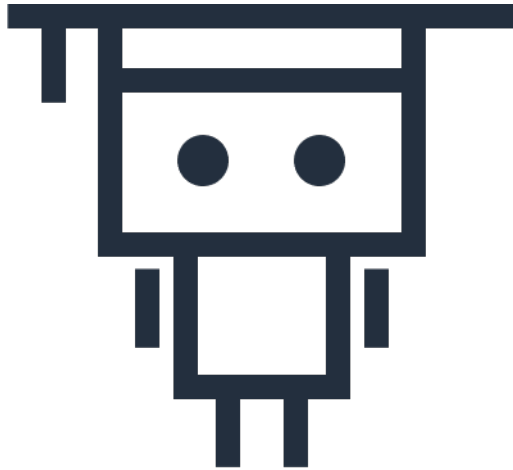
Image generation: Create realistic and artistic images of various subjects, environments, and scenes from language prompts.

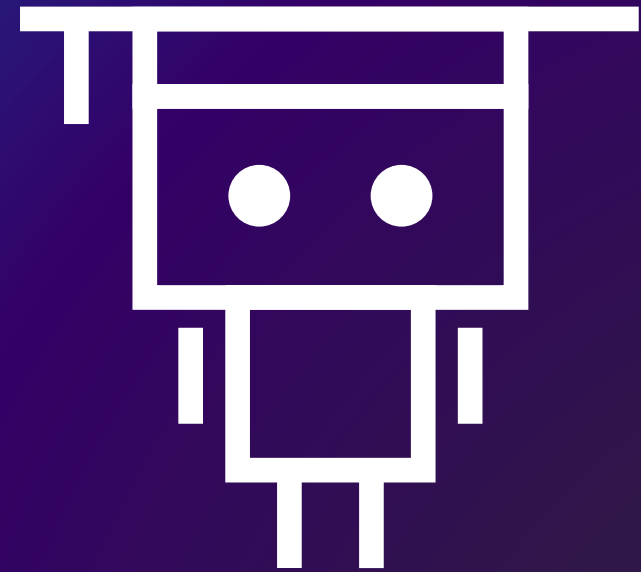


Personalization: Help customers find what they're looking for with more relevant and contextual product recommendations than word matching.

Next lesson

- This lesson covered fundamentals of generative AI
- In the next lesson, you will explore foundation models and large language models further





Thank you!